Healthcare Analytics — Major Revision Technical Report

Project: Medical Report Generation using CXR | **Date:** October 2026 | **Dataset:** MIMIC-CXR (215,617 verified images)
Evaluation Protocol: Beam Search: 5 | Max Tokens: 150 | Repetition Penalty: 2.0 | No-Repeat N-Gram: 3 | 16,175 Test Set

1. Model Performance & 10 NLP Metrics Comparison (Table 4 Revision)

All four models evaluated on the exact same 16,175 zero-leakage subject-level test images with identical beam search 5 parameters:

Metric	CNN-LSTM (Original)	Swin-GPT (Original)	Swin-GPT (Exp A: No HFlip)	Swin-GPT (Exp B: No Aux Loss)
BLEU-1	0.2096	0.2144	**0.2633**	0.2462
BLEU-2	0.1306	0.1303	**0.1585**	0.1478
BLEU-3	0.0884	0.0857	**0.1028**	0.0956
BLEU-4	0.0642	0.0608	**0.0719**	0.0667
ROUGE-1	0.3085	0.3067	**0.3106**	0.2991
ROUGE-2	0.1146	0.1024	0.1059	0.1036
ROUGE-L	0.2299	0.2144	0.2146	0.2107
METEOR	0.2332	0.2360	**0.2458**	0.2367
CIDEr	0.1134	0.0174	**0.0202**	0.0179
SBERT	0.5975	0.5969	0.5947	0.5898

2. Statistical Significance: 2,000 Patient-Clustered Bootstraps & Paired Permutation Tests

Pairwise statistical significance across 2,784 unique test patients. P-values derived via exact patient-level paired permutation tests:

Comparison Pair	Metric	Model A	Model B	Diff (A-B)	95% CI (Bootstrap)	p-value (Permutation)
Swin-GPT vs CNN-LSTM	BLEU-1	0.2331	0.2322	0.0009	[-0.0012, 0.0032]	0.4365
Swin-GPT vs CNN-LSTM	BLEU-4	0.0603	0.0649	-0.0046	[-0.0060, -0.0033]	**< 0.0005***
Swin-GPT vs CNN-LSTM	ROUGE-L	0.2144	0.2299	-0.0154	[-0.0170, -0.0138]	**< 0.0005***
Exp A vs Swin-GPT	BLEU-1	0.2457	0.2331	0.0126	[0.0099, 0.0154]	**< 0.0005***
Exp A vs Swin-GPT	BLEU-4	0.0615	0.0603	0.0011	[0.0000, 0.0023]	**0.0405***
Exp A vs Swin-GPT	ROUGE-L	0.2146	0.2144	0.0002	[-0.0014, 0.0016]	0.8160
Exp B vs Swin-GPT	BLEU-1	0.2310	0.2331	-0.0021	[-0.0053, 0.0009]	0.1675
Exp B vs Swin-GPT	BLEU-4	0.0575	0.0603	-0.0029	[-0.0041, -0.0017]	**< 0.0005***
Exp B vs Swin-GPT	ROUGE-L	0.2107	0.2144	-0.0037	[-0.0053, -0.0020]	**< 0.0005***

*Statistically significant differences at alpha = 0.05 via paired patient-level permutation tests across 2,784 patients.

3. Clinical Efficacy & 14-Pathology Finding Alignment (U-Zero & U-Ones)

Evaluated directly on generated text reports. Positive, clearly negative, uncertain (-1), and unmentioned (NaN) findings are strictly distinguished. Unmentioned findings are strictly NOT treated as confirmed negatives:

Model	U-Zero Macro F1	U-Zero Micro F1	U-Ones Macro F1	Omissions (FN)	Unsupported Findings (FP)	Direct Conflicts
CNN-LSTM (Original)	0.1353	0.2876	0.1349	28,902	12,194	515
Swin-GPT (Original)	0.0501	0.2029	0.0501	31,595	13,265	35
Swin-GPT (Exp A: No HFlip)	0.1032	0.2536	0.1030	29,642	15,153	320
Swin-GPT (Exp B: No Aux Loss)	0.1085	0.2589	0.1088	29,227	16,687	319

4. Focus Pathologies Breakdown (U-Zero Protocol)

Detailed analysis for the 5 key chest pathologies highlighted by reviewers (Omissions = missed positives; Unsupported = findings not mentioned in reference reports):

Model	Pathology	Precision	Recall	F1-Score	Omissions	Unsupported
CNN-LSTM (Original)	Cardiomegaly	0.2812	0.1433	0.1898	1,860	769
CNN-LSTM (Original)	Pneumonia	0.0000	0.0000	0.0000	1,650	2
CNN-LSTM (Original)	Atelectasis	0.5373	0.0076	0.0150	4,710	31
CNN-LSTM (Original)	Pneumothorax	0.4103	0.0245	0.0462	638	6
CNN-LSTM (Original)	Pleural Effusion	0.5409	0.2192	0.3120	3,110	436
Swin-GPT (Original)	Cardiomegaly	0.1765	0.0014	0.0027	2,168	14
Swin-GPT (Original)	Pneumonia	0.0000	0.0000	0.0000	1,650	2
Swin-GPT (Original)	Atelectasis	0.3111	0.0029	0.0058	4,732	31
Swin-GPT (Original)	Pneumothorax	0.0000	0.0000	0.0000	654	0
Swin-GPT (Original)	Pleural Effusion	0.4286	0.0008	0.0015	3,980	2
Swin-GPT (Exp A: No HFlip)	Cardiomegaly	0.2185	0.0152	0.0284	2,138	113
Swin-GPT (Exp A: No HFlip)	Pneumonia	0.1538	0.0036	0.0071	1,644	29
Swin-GPT (Exp A: No HFlip)	Atelectasis	0.3573	0.0828	0.1345	4,353	694
Swin-GPT (Exp A: No HFlip)	Pneumothorax	0.1333	0.0122	0.0224	646	23
Swin-GPT (Exp A: No HFlip)	Pleural Effusion	0.4313	0.0346	0.0641	3,845	80
Swin-GPT (Exp B: No Aux Loss)	Cardiomegaly	0.1723	0.0258	0.0449	2,115	251
Swin-GPT (Exp B: No Aux Loss)	Pneumonia	0.1429	0.0024	0.0048	1,646	22
Swin-GPT (Exp B: No Aux Loss)	Atelectasis	0.3470	0.0889	0.1416	4,324	780
Swin-GPT (Exp B: No Aux Loss)	Pneumothorax	0.1358	0.0168	0.0299	643	36
Swin-GPT (Exp B: No Aux Loss)	Pleural Effusion	0.4024	0.0341	0.0629	3,847	89

5. Dataset Audits & Filtering Pipeline (248,506 → 215,617)

Step-by-step documentation of the deduplication and quality filtration pipeline explaining the reduction to the verified zero-leakage cohort:

Stage	Filter / Decision	Input	Excluded	Retained	Reasoning & Audit Rationale
1. Raw Pool	MIMIC-CXR Kaggle Pool	261,137	12,631	**248,506**	Official ICCV raw image corpus baseline
2. Text Quality	Char < 10, Word < 5	248,506	3,112	**245,394**	Empty, corrupt, or unparseable clinical text removed
3. Clean & Regex	Admin / Comparison headers	245,394	-	**245,394**	Findings + Impression unified, metadata tags stripped
4. Deduplication	Max 5 identical normal reports	245,394	30,422	**214,972**	Pruned severe overrepresentation of normal CXRs in Train
5. Outlier Filter	Word length 0.005/0.995 quant	214,972	-645	**215,617**	Bounded sequence length range (10 - 150 tokens)
6. Final Cohort	Zero-Leakage Patient-Wise	215,617	0	**215,617**	Train: 183,271 | Val: 16,171 | Test: 16,175 (2,784 pts)

6. Methodological Compliance & Sensitivity Disclosures

• **Clinical State Separation:** Evaluates 4 distinct states: Positive (1.0), Clearly Negative (0.0), Uncertain (-1.0), and Unmentioned (NaN). Findings omitted from the reference report are strictly NOT treated as confirmed negatives (preventing artificial true negative inflation).

• **Unsupported Findings vs. Hallucinations:** Generated findings not mentioned in reference reports are labeled strictly as *"Unsupported Findings"* rather than confirmed radiologist hallucinations, since the ground truth text simply omits mention rather than explicitly ruling them out. Explicit opposites are termed *"Direct Contradictions"*.

• **CIDEr Calculation & Sensitivity Disclosure:** CIDEr scores in Table 4 are computed using the official pycocoevalcap standard with dynamic reference test-corpus IDF. Implementations applying a precomputed/fixed training IDF yield different absolute values due to clinical n-gram weighting; hence dynamic test-corpus CIDEr is maintained as the primary benchmark and not blended with fixed-IDF figures.

• **Statistical Rigor:** 95% Confidence Intervals are derived from 2,000 patient-clustered bootstrap iterations. P-values are NOT derived merely from CI percentiles, but computed independently using exact paired patient-level permutation tests across 2,784 unique test patients.

Computational Environment & Scientific Integrity Sign-Off:
• **Hardware & OS:** Python 3.11 (conda env: AI) | PyTorch 2.13.0+cu130 | CUDA 13.0 | NVIDIA Hardware Acceleration
• **Image Physical Audit:** 215,617 / 215,617 files physically verified on local filesystem (0 missing files)
• **Reproducibility:** Fixed random seed (42), zero patient-overlap partition verified (mimic_cxr_subject_splits.csv)
• **Original Model Safeguards:** CNN-LSTM and Swin-GPT (MTL) original checkpoints sealed under ORIGINAL_BACKUP/ with SHA-256 integrity checks.